

INSTRUCTIONS

- The deadline for the quiz is **Friday, 14 March 2025, 23:59**.
- Submit your completed quiz as a single PDF document through Ulwazi. Please do not email your submissions.
- Submit your code via the link provided on Ulwazi (separate from the PDF).
- Collaborate in groups of *three to five* members. Only one submission per group is necessary.
- The quiz consists of a total of 5 questions, with an overall allocation of 20 marks.
- Full solutions must be provided to receive full marks. Simply providing the correct answer without showing your work will not suffice.

QUESTIONS

1. Consider the feasible region defined by

$$S = \{\mathbf{x} \in \mathbb{R}^2 : x_2 - (1 - x_1)^3 \leq 0, \quad 2x_1 + x_2 \leq 2\}.$$

- (a) Verify whether the Linear Independence Constraint Qualification (LICQ) holds at the optimal point $\mathbf{x}^* = [1, 0]^T$. [2]
- (b) Discuss the significance of LICQ in the context of the Karush-Kuhn-Tucker (KKT) conditions and why it is necessary for ensuring optimality. [2]

2. Consider the following constrained optimization problem:

$$\begin{aligned} &\text{minimize } f(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 1)^2 \\ &\text{subject to : } -\frac{1}{4}x_1^2 - x_2^2 + 1 \geq 0 \\ &\quad \quad \quad x_1 - 2x_2 + 1 = 0. \end{aligned}$$

Find the solution(s) of the problem by applying the Karush-Kuhn-Tucker (KKT) and second-order optimality conditions. [4]

3. Consider the following problem:

$$\min \frac{1}{2} (x_1^2 + x_2^2) \quad \text{subject to } 2 - x_1 - x_2 = 0.$$

- (a) Calculate the minimum point of the quadratic penalty function as a function of the penalty parameter. (i) Show that $\mathbf{x}(\mu) \rightarrow \mathbf{x}^*$ as $\mu \rightarrow \infty$, and (ii) show that $-\mu c(\mathbf{x}(\mu)) \rightarrow \lambda^*$ as $\mu \rightarrow \infty$ (note $c(\cdot)$ is the constraint). [2]

- (b) Calculate the Hessian of the quadratic penalty function, and show that it becomes ill-conditioned as penalty parameter $\mu \rightarrow \infty$. [2]

4. Consider the constrained optimization problem

$$\min_{\mathbf{x}} f(\mathbf{x}) = \frac{1}{2}x_1^2 - x_1x_2 \quad \text{subject to } x_1 + x_2 = 1.$$

Calculate the minimum point as a function of the Lagrangian multiplier λ and the penalty parameter μ . Find the update of the Lagrangian multiplier in terms of the penalty parameter. Find the value of the minimizer and optimal Lagrange multiplier as $\mu \rightarrow \infty$. [4]

5. Consider the following optimization problem, minimize $f(x, y, z) = xy + xz + yz$ subject to the constraints: $x^2 + y^2 + z^2 = 2$ and $z = 1$.

Write a Python program that implements the Augmented Lagrangian Method to find the minimizer and the corresponding Lagrange multipliers for this problem. Ensure that the code runs smoothly and is well-documented. Your implementation will be tested on various initial conditions and parameters. [4]

***** Good luck! *****